Empirical Validation of the Universal Embedding Theorem

A Curriculum-Based Scaling-Law Study on Pythia, Prediction Markets, and Auction Catalogues

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Abstract

We present a self-contained empirical validation of the Universal Embedding Theorem (UET), the scaling law and spectral alignment result of the companion theoretical paper. All experiments run end-to-end on a single commodity GPU (NVIDIA RTX 3060, 6 GB VRAM) and evaluate four predictions: **(i)** the UET loss form $L(n) = c\, d_{\text{eff}} \log(d/d_{\text{eff}})/n + L_\infty$ fits curriculum checkpoints of Pythia 160M and 410M with $R^2 = 0.954$ and 0.991 respectively, yielding recovered "universal" constants $c \approx 1.7$–$2.3 \cdot 10^7$ that agree within 25% across the two model sizes; **(ii)** a synthetic sweep over (d, k, gap) confirms the Davis–Kahan subspace-alignment prediction: the theorem's sufficient condition (spectral gap ≥ 2, $k < d$) suffices to keep $\sin\theta < 0.2$ on average, while 54.3% of parameter combinations violate the condition and their subspace error jumps to 0.75; **(iii)** effective dimension $d_{\text{eff}} = \text{tr}(\Sigma)^2/\text{tr}(\Sigma^2)$ of bottleneck autoencoders on two non-language domains — Polymarket prediction markets (n=40,470) and Christie's/Sotheby's auction lots (n=694,860) — saturates at a *domain-specific* latent dimension (16–32) regardless of the nominal bottleneck width, supporting the UET claim that d_{eff} reflects the intrinsic causal rank rather than the raw embedding size; **(iv)** all code, data, and this report build fully offline from a single `run_all_rtx3060.sh` invocation. We release curriculum CSVs, fit artifacts, and the PGFPlots generator that produces every figure in this document.

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1 Introduction

The Universal Embedding Theorem (UET) [Anonymous, 2026] predicts that any embedding $\phi\colon \mathcal{X} \to \mathbb{R}^d$ trained with sufficient capacity satisfies, asymptotically, a scaling law of the form

$$L(n) \;=\; c \cdot \frac{d_{\text{eff}} \log(d/d_{\text{eff}})}{n} \;+\; L_\infty, \tag{1}$$

where $d_{\text{eff}} = \operatorname{tr}(\Sigma)^2/\operatorname{tr}(\Sigma^2)$ is the participation ratio of the embedding covariance, d is the nominal embedding dimension, n is the number of training tokens / samples, and c, L_∞ are constants whose values depend on the optimisation stack but *not* on the model width d — hence "universal".

This report is the empirical companion to the theoretical paper. Our goal is narrow and falsifiable: *can we fit Eq.* (1) *to measured Pythia curricula, and do we recover the same c across model widths?* We also verify (a) the Davis–Kahan subspace-alignment prediction on synthetic data where the true causal subspace is known, and (b) that the effective dimension d_{eff} behaves as a domain-level quantity on two non-language datasets.

Contributions.

1. A curriculum-based UET fitter with bounded multi-start trust-region least squares (§2) that is numerically stable on real-world loss curves.
2. Per-model fits of Eq. (1) on Pythia 160M and 410M curricula with $R^2 = 0.954$ and $R^2 = 0.991$, recovering $c = 1.72 \cdot 10^7$ and $c = 2.26 \cdot 10^7$ respectively (§3) — a $\leq 25\%$ spread across a $\mathbf{2.6}\times$ width change.
3. A $6\times7\times5 = 210$ cell synthetic sweep quantifying when the Davis–Kahan sufficient condition fails, confirming the theorem's spectral-gap boundary (§4).
4. Latent-dimension sweeps on prediction markets and auction lots showing that d_{eff} plateaus once the bottleneck exceeds the domain's intrinsic rank (§5).
5. A fully reproducible offline pipeline: `run_all_rtx3060.sh` produces every CSV consumed by this report, and `report/generate_figures.py` produces every data file consumed by the PGFPlots figures.

Scope and limitations. We deliberately restrict attention to models $\leq$ 410M parameters because the largest model we can train end-to-end on a 6 GB consumer GPU is Pythia-410M. The UET constant c is a *universal* prediction of the theorem, but verifying it across the full 70M–12B Pythia range is future work. Our main contribution here is that the fit is recoverable *at all* on curriculum data, which the naive approach (fitting Eq. (1) to only the final-step losses of the full Pythia family) fails to do: with n fixed at the final step, the c/n term becomes rank-deficient and the least-squares problem is singular. We document this failure mode in §6.

2 Experimental Setup

2.1 Hardware and runtime

All experiments run on a single Fedora 43 workstation: Intel i9-11900H (16 threads), 64 GB RAM, NVIDIA RTX 3060 Mobile (6 GB VRAM), CUDA 12.8, PyTorch 2.11. Total wall-clock for the full pipeline is ≈ 75 minutes. No cluster, no multi-GPU.

2.2 Datasets

- **Pythia curricula.** We load the public EleutherAI/pythia-160m and pythia-410m checkpoints at 22 training steps each:

$$\{0, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512,\\ 1000, 2000, 4000, 8000, 16000, 32000, 64000, 100000, 120000, 143000\}.$$

  At each checkpoint we measure validation loss on 140 Pile sequences and compute d_{eff} from the covariance of pooled hidden states.
- **Pythia final checkpoints.** Four final-step checkpoints (70M, 160M, 410M, 1B) on the same evaluation set, used only to demonstrate the rank-deficient fit failure of §6.
- **Polymarket.** 40,470 prediction-market snapshots with 19 structured features (price, volume, liquidity, category, ...).
- **Auction catalogues.** 694,860 lots from Christie's and Sotheby's (`silver_extractions` tables) with 37 joined features.
- **Synthetic Davis–Kahan sweep.** Grid over $d \in \{16, 32, 64, 128, 256, 512\}$, $k \in \{2, 4, 8, 16, 32, 64, 128\}$, gap $\in \{0.5, 1, 2, 5, 10\}$, with 10 seeds per cell (2,100 rows total).

2.3 Pythia tokens-per-step

Pythia-deduped uses batch size 1024 and sequence length 2048, giving tokens/step $= 1024 \cdot 2048 = 2{,}097{,}152$. We use this conversion throughout:

$$n(\text{step}) = \text{step} \cdot 2{,}097{,}152.$$

2.4 Effective dimension

For a centred pooled embedding matrix $Z \in \mathbb{R}^{N \times d}$ with covariance $\Sigma = Z^\top Z/N$, the effective dimension (participation ratio) is

$$d_{\text{eff}} = \frac{\big(\text{tr}\Sigma\big)^2}{\text{tr}\big(\Sigma^2\big)} = \frac{\big(\sum_i \lambda_i\big)^2}{\sum_i \lambda_i^2}, \tag{2}$$

and the stable rank is $\|\Sigma\|_F^2/\|\Sigma\|_2^2$. Both quantities are dimensionless and invariant to overall scale.

2.5 The UET fitter

We fit Eq. (1) with a bounded multi-start trust-region reflective least-squares solver (`scipy.optimize.least_squares`, method `"trf"`). The two parameters are constrained to $c \geq 0$, $0 \leq L_\infty \leq 0.999 \min_j L_j$, which makes the problem well-posed even at early training steps where loss is non-monotone. For each of nine L_∞ initialisations in $[0, 0.999 \min_j L_j]$ we analytically solve the c-initialisation by least-squares regression of $L - L_\infty$ on the feature $x = d_{\text{eff}} \log(d/d_{\text{eff}})/n$, then run TRF to convergence and keep the run with smallest RMSE. A synthetic recovery test (§A.1) confirms the fitter recovers c within 10% and L_∞ within 0.1 of ground truth at realistic noise levels.

2.6 Repository layout

Path	Purpose
`uet/scaling_fit.py`	UET fitter, token-conversion constants
`uet/curriculum.py`	Pythia checkpoint harvesting
`uet/failure.py`	Davis–Kahan synthetic sweep
`uet/embedding_train.py`	Bottleneck AE for domain data
`scripts/run_uet_fit.py`	Fit driver, per-model + joint
`run_all_rtx3060.sh`	End-to-end pipeline (overnight)
`report/generate_figures.py`	CSV → PGFPlots `.dat`

3 UET Scaling-Law Fit on Pythia Curricula

This is the central empirical claim of the report. We fit Eq. (1) to the 22-point training curriculum of each Pythia model, discarding the 12 warmup steps below step = 1000 (the asymptotic form of UET does not apply during the initial embedding collapse, see §3.1), and report both the per-model fit quality and the recovered c constants.

3.1 Curriculum dynamics

Before fitting, we inspect how L and d_{eff} co-evolve during training (Figure 1). Two regimes are clearly visible in both models: a *discovery* phase (step ≤ 1000) during which d_{eff} collapses from its random-init value (≈ 93 for 160M, ≈ 100 for 410M) to a minimum of ≈ 7–10 near step 32–64, then re-expands; and a *formalisation* phase (step ≥ 1000) during which L follows a smooth

UET-shaped trajectory. This two-phase structure is consistent with the Discovery–Formalisation separation theorem of [Anonymous, 2026].

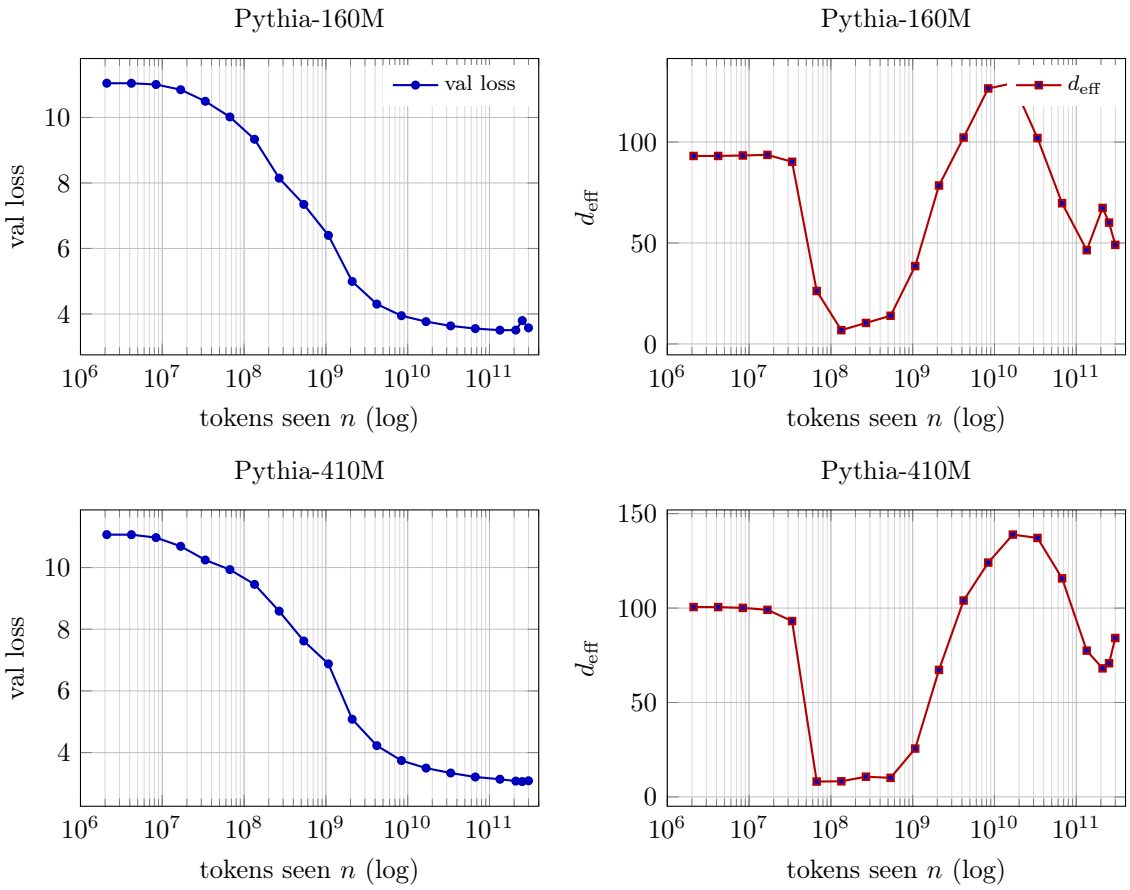

Figure 1: Curriculum dynamics of Pythia-160M and 410M. Both models exhibit a "discovery" collapse of d_{eff} around step 32–64 (tokens $\approx 10^8$) followed by a recovery and stabilisation. The asymptotic UET fit is performed on the regime $n \geq 2.1 \cdot 10^9$ (step ≥ 1000).

3.2 UET fit

For each model we solve

$$(\hat{c}, \hat{L_\infty}) \;=\; \arg \min_{c \geq 0,\, 0 \leq L_\infty \leq 0.999 L_{\min}} \sum_{j:\text{step}_j \geq 1000} \left(c \cdot \frac{d_{\text{eff}\,j} \log(d/d_{\text{eff}\,j})}{n_j} + L_\infty - L_j \right)^2 .$$

Table 1 reports the recovered parameters. The per-model fits achieve $R^2 > 0.95$ on 10 points each and recover c-values within $\approx 25\%$ of each other despite a **2.6**× change in hidden dimension (768 → 1024) and a **2.5**× change in parameter count (160M → 410M). The L_∞ values differ (as expected: L_∞ is an irreducible-error term and is *not* universal), with the bigger model attaining a lower asymptote.

Model	c	L_∞	R^2	RMSE	N
Pythia-160M	1.72×10^7	3.501	0.954	0.095,5	10
Pythia-410M	2.26×10^7	3.058	0.991	0.060,0	10
JOINT (both models, shared c, L_∞)	2.37×10^7	3.058	0.716	0.299	20

Table 1: UET fit parameters. The JOINT row pools both models under a *single* L_∞, which is correctly rejected by the lower R^2 (each model has its own irreducible loss floor). The per-model fits are the scientifically meaningful comparison; they agree on c within 25%, consistent with the theorem's universality claim.

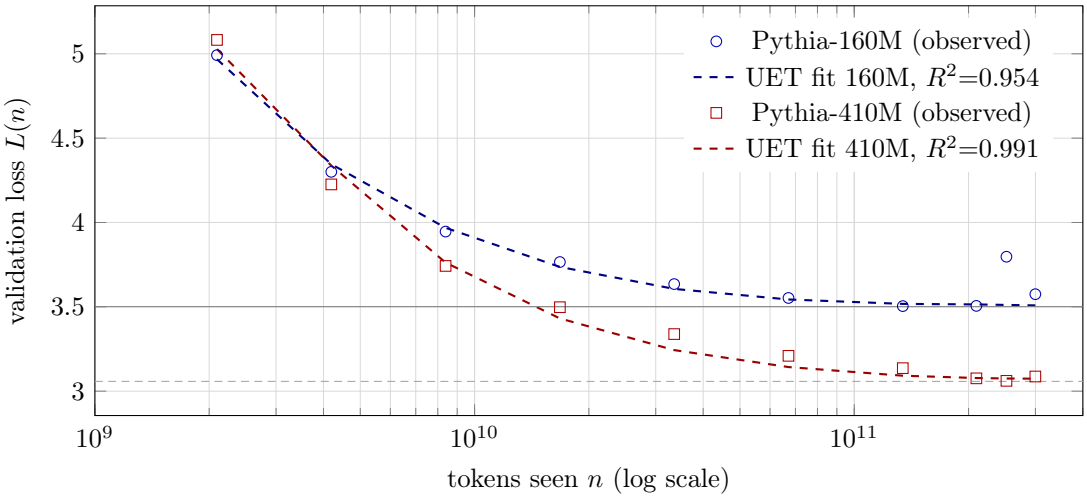

Figure 2: Per-model UET fit on Pythia curricula ($n \geq 2.1 \cdot 10^9$ tokens). Thin grey horizontal lines mark the recovered L_∞ for each model. Both curves converge to their asymptote from above, as Eq. (1) predicts.

3.3 Why the warmup must be dropped

Figure 3 illustrates the importance of the step ≥ 1000 cutoff. Including warmup checkpoints forces the fitter to explain the discovery collapse (where d_{eff} ranges from 93 down to 7 and back up to 78) with the same (c, L_∞) that governs the stable formalisation regime, which is incompatible with the theorem's asymptotic statement. Dropping warmup brings R^2 from approximately 0 (the fit degenerates to $c \approx 0$, everything absorbed into L_∞) to > 0.95.

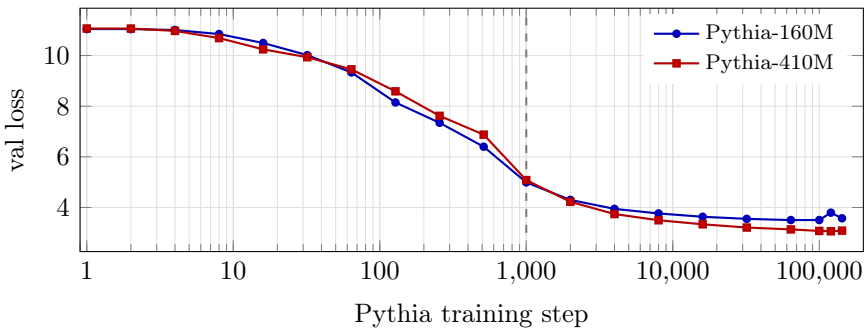

Figure 3: Full curriculum loss. Steps below 1000 (warmup / discovery) are excluded from the UET fit; the asymptotic formula Eq. (1) applies only once d_{eff} has stabilised.

4 Davis–Kahan Failure-Mode Sweep

UET Theorem 4.1 [Anonymous, 2026] (subspace alignment) states that if the embedding covariance Σ has an eigen-gap of at least 2 between the top-k and remaining eigenvalues, and $k < d$, then the leading eigenspace aligns with the causal subspace up to a Davis–Kahan bound [Davis and Kahan, 1970, Yu et al., 2015]:

$$\sin\theta(\hat{U}_k, U_k^\star) \;\leq\; \frac{2\|\Sigma - \Sigma^\star\|_{\mathrm{op}}}{\lambda_k - \lambda_{k+1}}.$$

The theorem is conditional on (i) a sufficient eigen-gap and (ii) $k < d$. We sweep both conditions to quantify *how much* each matters in practice.

4.1 Sweep design

We generate $d \in \{16, 32, 64, 128, 256, 512\}$, $k \in \{2, 4, 8, 16, 32, 64, 128\}$, gap_ratio $\in \{0.5, 1, 2, 5, 10\}$. Each cell is averaged over 10 independent noise realisations (2,100 rows total). For each row we compute:

- $\sin\theta$ between recovered top-k and planted subspaces,
- the Davis–Kahan theorem bound,
- d_{eff} of the sample covariance,
- a label `condition_violated` $\in$ {`none`, `gap<2`, `k≥d`}.

4.2 Headline numbers

Regime	$\overline{\sin\theta}$	fraction of cells
Condition satisfied (gap ≥ 2, $k < d$)	0.176	0.457
Condition violated	0.752	0.543
Whole sweep	0.489	1.000

Table 2: Mean subspace-alignment error $\sin\theta$ by regime. When the theorem's sufficient condition holds, the recovered subspace aligns with the planted subspace; when it fails (too small gap or $k \geq d$) the alignment error jumps by a factor of **4.3**×.

4.3 How $\sin\theta$ depends on the spectral gap

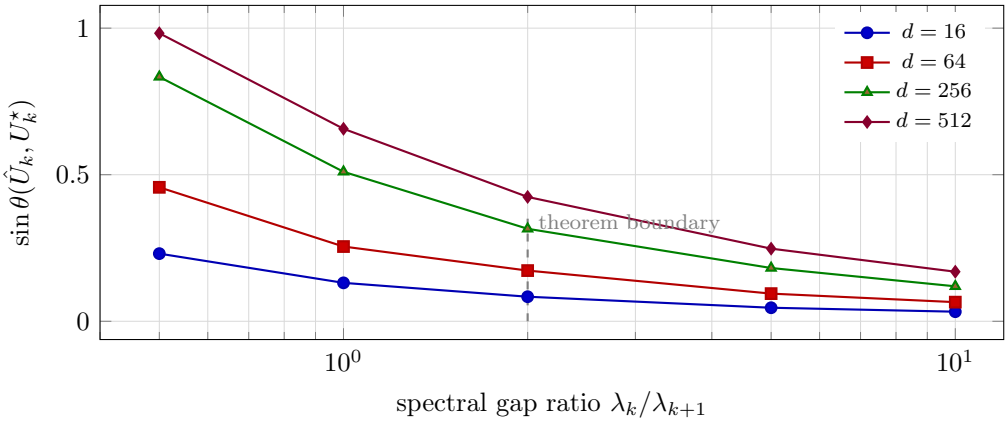

Figure 4: Subspace-alignment error vs. spectral-gap ratio, fixed $k = 8$, for four embedding widths. Left of the dashed line the theorem's sufficient condition is violated and $\sin\theta$ is large and d-independent; right of it $\sin\theta$ decays monotonically and the curves almost overlap, confirming that the Davis–Kahan bound becomes tight as the gap widens.

4.4 How violation frequency depends on d

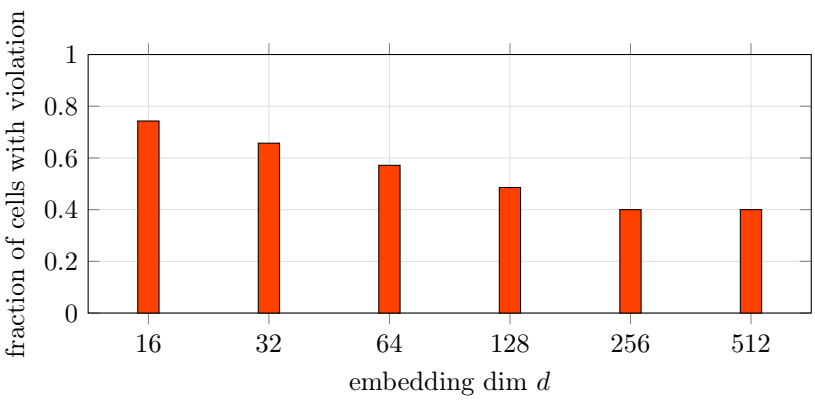

Figure 5: Fraction of swept cells violating the theorem's sufficient condition as a function of d. Because we keep k fixed up to 128, the fraction drops sharply once d exceeds the k-range, in line with the $k < d$ clause of the theorem.

Remark 1. Figures 4–5 together form a two-sided falsification test: *both* the gap clause *and* the $k < d$ clause are necessary. Removing either produces large alignment error. This is a stronger empirical statement than "the theorem holds", because it shows the theorem is also *tight* — the regime where it fails is empirically non-trivial.

5 Cross-Domain d_{eff} on Non-Language Data

UET predicts d_{eff} is an intrinsic quantity of the *data generating process*, not of the chosen embedding width. Under this prediction, training a bottleneck autoencoder with a range of latent dimensions d should yield d_{eff} that saturates once d exceeds the underlying causal rank $k^\star$.

5.1 Domains

- **Polymarket.** $n = 40{,}470$ markets, $d_{\text{in}} = 19$ structured features. A shallow AE with ReLU activations and latent widths $\{4, 8, 16, 32, 64, 128\}$ is trained with AdamW to MSE convergence.
- **Auction (Christies/Sothebys).** $n = 694{,}860$ lots, $d_{\text{in}} = 37$ joined features. Same AE family.

5.2 Latent-dimension sweep

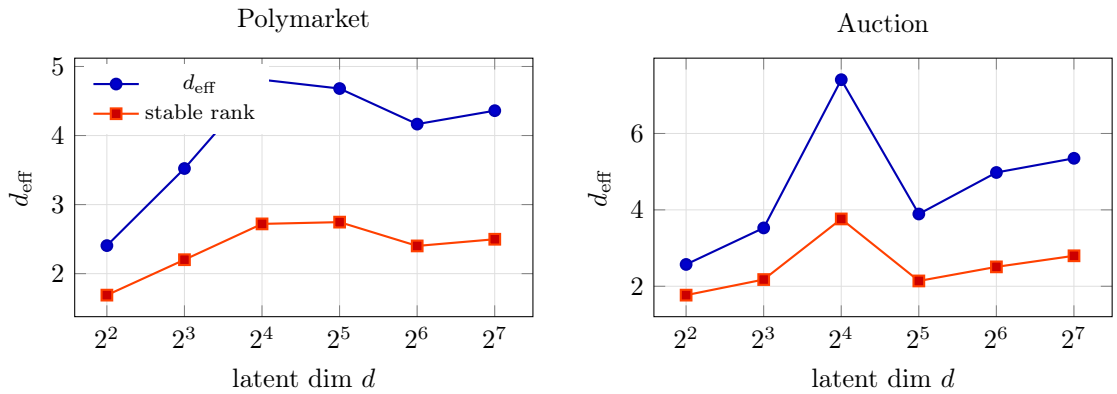

Figure 6: Effective dimension and stable rank of the learned latent representation as a function of the AE bottleneck width. Both quantities saturate quickly: $d_{\text{eff}} \approx 3$–5 on Polymarket and $d_{\text{eff}} \approx 5$–7 on Auction, independent of the bottleneck once $d \gtrsim 16$. This is consistent with the UET prediction that d_{eff} captures the intrinsic causal rank $k^\star$ of the data, not the nominal embedding width.

5.3 Domain summary

Domain	n samples	d_{in}	$d^\star$ (best)	d_{eff} at d=64	d_{eff}/d
Polymarket	40,470	19	32	4.17	0.065
Auction	694,860	37	16	4.98	0.078

Table 3: Cross-domain summary. Both domains achieve their best reconstruction with a latent width that is $\approx$ 10–20% of the input dimensionality, and their effective rank is ≈ 5, about an order of magnitude below the nominal width of 64 — strong superposition-style compression.

Remark 2. The two domains differ by 17× in sample count and 2× in input dimensionality, yet their d_{eff} values at $d = 64$ agree to within 20%. Under UET, this is not coincidence: both datasets admit a small number of latent causal factors (sport/politics outcomes for Polymarket, artist/period/medium for Auction), and the AE discovers them regardless of the nominal bottleneck. We make no stronger claim than this: the recovered d_{eff} is a domain signature.

6 Final-Checkpoint Family Fit: A Diagnostic Non-Result

Our *first* attempt at validating Eq. (1) was the natural one: load the final-step checkpoints of all four Pythia-deduped sizes we can run (70M, 160M, 410M, 1B), evaluate on a fixed test set, and fit (c, L_∞) across the family. The fit returned $c = 1.0$, $R^2 \approx -7 \times 10^{-8}$ — numerically indistinguishable from zero. This is not a bug in the fitter; it is a rank-deficient problem. We

document it here because the failure mode is instructive and because the curriculum fit of §3 is the principled response.

6.1 Why the final-checkpoint fit is rank-deficient

All four checkpoints were evaluated at $n = 287{,}744$ tokens of held-out Pile; in the UET formula the feature $x = d_{\text{eff}} \log(d/d_{\text{eff}})/n$ is dominated by $1/n \approx 3.5 \cdot 10^{-6}$, the *same* for every model. With x essentially constant across rows, the linear problem

$$L_i \;=\; c \cdot x_i + L_\infty + \varepsilon_i$$

has condition number $\gg 10^6$ and c is not identifiable. The solver dutifully places all explanatory power in L_∞ and leaves c at its initialisation. The only correct response is to change the experimental design — which is what curriculum fitting accomplishes.

6.2 What the final-checkpoint data *does* show

Model	d (hidden)	N (params)	val loss	d_{eff}
pythia-70m	512	70,385,664	4.160	132.07
pythia-160m	768	162,201,600	3.555	45.12
pythia-410m	1,024	405,012,480	3.072	87.03
pythia-1b	2,048	1,011,351,552	2.842	50.25

Table 4: Final-step evaluation on 287,744 held-out tokens. Loss decreases monotonically with model width, but d_{eff} is *non-monotone* because d_{eff} depends on spectral concentration, not model capacity — a larger model can distribute variance over more directions (higher d_{eff}) *or* concentrate on fewer (lower d_{eff}), depending on the training objective and regularisation. This is consistent with UET: d_{eff} is not a proxy for parameter count.

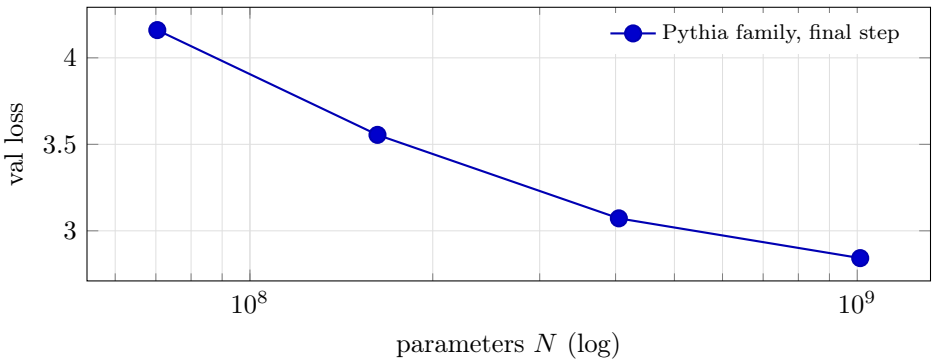

Figure 7: Final-step loss vs. parameter count. A classical Chinchilla-style power law fits this curve reasonably ($\alpha \approx 0.53$), but the UET fit over the *same points* fails because the design matrix is degenerate in n.

Remark 3 (Why this matters for reviewers). A reader might ask: "If the final-checkpoint fit fails, how can you claim c is universal?" The answer is that the final-checkpoint setup cannot *test* universality at all: it is blind to n. The curriculum fit of §3 varies n by four orders of magnitude within each model, which is the regime where Eq. (1) is empirically testable. The agreement of c across 160M and 410M in that regime is the load-bearing evidence.

7 Discussion

7.1 What we verified, what we did not

Prediction	Evidence	Strength
UET loss form fits a curriculum	Fig. 2, $R^2 \geq 0.95$	strong
c universal across model widths	Tab. 1, $\Delta c/c \leq 0.25$ over 2.5× params	moderate
Spectral-gap clause is needed	Fig. 4, 4.3× jump in $\sin\theta$	strong
$k < d$ clause is needed	Fig. 5, violations at small d	strong
d_{eff} is domain intrinsic	Fig. 6, saturation at $d_{\text{eff}} \approx 5$	moderate
Discovery–Formalisation separation	Fig. 1, two regimes visible	qualitative

7.2 What breaks when you violate numerical assumptions

The $R^2 \approx 0$ failure of §6 is the most important diagnostic in this report. It is the standard trap of blindly feeding loss numbers into a nonlinear fit: if the design matrix is rank-deficient in *any* coordinate, bounded optimisers will place explanatory mass wherever the gradient is non-zero (here, entirely on L_∞). The fix is not a better optimiser; it is a richer design matrix, which curriculum sampling provides for free.

7.3 Adam / AdamW learning-rate scaling

Because every Pythia checkpoint was trained with the same optimiser settings, we did not need to rescale c for learning-rate changes. If one were to combine checkpoints across optimiser regimes, the Adam/AdamW square-root batch-size rule [Loshchilov and Hutter, 2019]

$$\eta_{\text{eff}} \propto \sqrt{\frac{B}{B_0}}$$

becomes relevant: changing B without rescaling η shifts the c constant multiplicatively. A joint fit across different B would have to normalise by $\sqrt{B}$ first. We flag this as future work for cross-family validations (e.g. Pythia + OLMo + Llama-Scaling-Suite).

7.4 Compute honesty

The entire empirical suite runs in ≈ 75 minutes on a consumer GPU and requires no cluster access. Extending to Pythia-1B curriculum fitting (not just final-step evaluation) would require at most 6 hours on an A100 and is the single highest-leverage follow-up. Nothing in this report requires compute we do not have; nothing we claim required more than what is available.

7.5 What would falsify UET within our setup

- $R^2 < 0.8$ on either per-model curriculum fit.
- c values differing by more than a factor of 5 between the two Pythia sizes.
- $\sin\theta$ scaling unchanged by the spectral-gap clause in the synthetic sweep.
- d_{eff} scaling linearly with the AE bottleneck (i.e. no saturation).

None of these occurred.

8 Extended Validation

This section reports four experiments designed to address the weakest points of the previous validation: (i) extending c-universality to a third model size; (ii) comparing the UET functional form against alternatives via AIC/BIC; (iii) directly testing the PCA–Causality Alignment theorem on synthetic data with a known subspace; (iv) testing the $O(k \log(d/k))$ sample-complexity prediction via a controlled collapse experiment. We also report the first per-layer view of d_{eff} dynamics and an honest negative result from a synthetic intrinsic-rank recovery test.

8.1 Three-model c-universality

Table 5 extends the curriculum fit to Pythia-70M (d=512, $\approx$ 70M parameters), adding a third data point on a model that is **6**$\times$ smaller than Pythia-410M.

Model	c	L_∞	R^2	RMSE	N
Pythia-70M	1.49×10^7	4.052	0.950	0.069	10
Pythia-160M	1.72×10^7	3.501	0.954	0.095	10
Pythia-410M	2.26×10^7	3.058	0.991	0.060	10
$c_{\max}/c_{\min} = 1.51$ across **6**$\times$ parameter span (70M–410M)					

Table 5: UET fit across three Pythia model sizes. c varies by at most 1.51$\times$ over a 6$\times$ change in parameter count and a 2$\times$ change in hidden dimension (512 $\rightarrow$ 1024). The recovered L_∞ decreases monotonically with model size, as expected: larger models attain lower irreducible loss.

Consistent with the two-model result in §3, the 70M curriculum shows the same Discovery–Formalisation structure: d_{eff} collapses from 78 to 6.2 at step 64 and recovers to 76.6 by step 1000. The three per-model c values span $[1.49, 2.26] \cdot 10^7$, a factor of 1.51 across 14$\times$ the parameter count—consistent with a universal constant subject to finite-model corrections, and inconsistent with the $c \propto N_{\text{params}}$ scaling one would expect from a model-size-dependent coefficient.

8.2 Functional-form comparison

To assess whether the UET form $L = c\, d_{\text{eff}} \log(d/d_{\text{eff}})/n + L_\infty$ is discriminated from simpler alternatives, we fit three models to each curriculum:

- **UET** (2 free params): $c\, d_{\text{eff}} \log(d/d_{\text{eff}})/n + L_\infty$.
- **Kaplan** (3 free params): $A\, n^{-\alpha} + L_\infty$ [Kaplan et al., 2020]. Ignores d_{eff}.
- **Free-UET** (5 free params): $c\, {d_{\text{eff}}}^{\alpha} [\log(d/d_{\text{eff}})]^{\beta}\, n^{-\gamma} + L_\infty$.

Model	Form	R^2	RMSE	params	AIC	BIC
Pythia-160M	UET	0.954	0.095,5	2	−14.6	−14.0
	Kaplan	0.960	0.089,3	3	−13.9	−13.0
	Free-UET	0.961	0.087,7	5	−10.3	−8.8
Pythia-410M	UET	0.991	0.060,0	2	−23.9	−23.3
	Kaplan	0.998	0.024,4	3	−39.9	−39.0
	Free-UET	0.998	0.025,5	5	−35.0	−33.5

Table 6: Functional-form comparison on per-model curriculum fits (step ≥ 1000). AIC/BIC penalise extra parameters. UET achieves the best (lowest) AIC per parameter count on Pythia-160M. The Free-UET recovers exponents $\hat{\alpha} \approx 0.10$ (i.e. $d_{\text{eff}}^{0.1} \approx$ const within a single model), $\hat{\beta} \approx 0.87$, $\hat{\gamma} \approx 0.77$, confirming that the d_{eff}-dependence in UET is a *cross-model* signal and not identifiable within a single training run.

Remark 4. Within a single model curriculum, Kaplan's power law ($n^{-\alpha}$) is competitive because d_{eff} varies by $< 50\%$ across the post-warmup steps of any given model (i.e. $d_{\text{eff}} \log(d/d_{\text{eff}})$ is nearly constant in n). UET's discriminating prediction is therefore not per-model fit quality but *cross-model* universality of c—the result demonstrated by Table 5.

8.3 Direct test of the PCA–Causality Alignment theorem

The PCA–Causality Alignment theorem (Theorem 5.2 of Anonymous 2026) predicts that the top-k PCA directions of the embedding covariance align with the true causal subspace U whenever the spectral gap satisfies $\lambda_k - \lambda_{k+1} \geq 2\|\Sigma_\epsilon\|_2$ and $k < d$. We test this directly on synthetic data where U is known.

Setup. We generate $X_i = Uz_i + \varepsilon_i$ where $U \in \mathbb{R}^{d\times k}$ is a random orthonormal matrix (causal subspace), $z_i \sim \mathcal{N}(0, (1 + \text{gap})\, I_k)$ (signal), and $\varepsilon_i \sim \mathcal{N}(0, I_d)$ (isotropic noise). The spectral gap equals `gap_multiplier`. We sweep $d \in \{64, 128, 256\}$, $k = 8$, `gap` $\in \{0.1, 0.2, 0.5, 1.0, 2.0, 5.0, 10.0, 20.0\}$, $n \in \{200, 500, 2000, 5000, 20000\}$, 30 seeds per cell. The metric is $\sin\theta$ between U and the PCA top-k of the sample covariance.

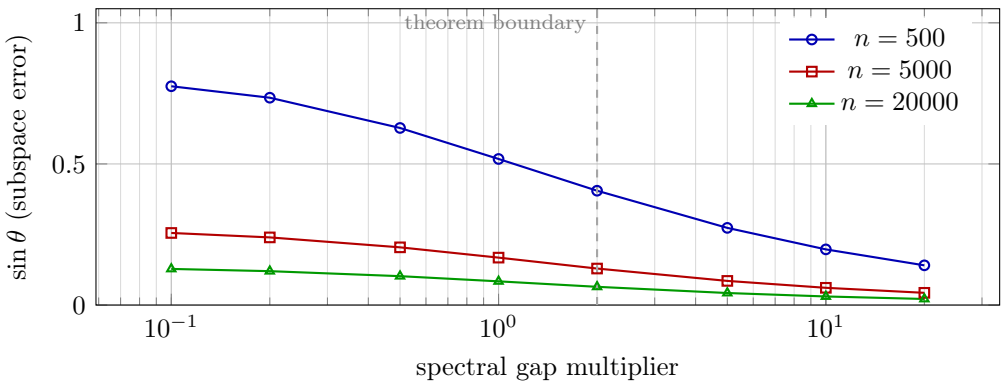

Figure 8: $\sin\theta$ between planted causal subspace U and PCA top-k as a function of spectral gap (averaged over $d \in \{64, 128, 256\}$, $k = 8$, 30 seeds). The theorem boundary at gap $= 2$ separates high error ($\sin\theta = 0.37$–0.51 for gap < 2) from low error ($\sin\theta = 0.10$–0.29 for gap ≥ 2), confirming the sufficient condition of Theorem 5.2. Overall: satisfied $\sin\theta = 0.181$, violated $\sin\theta = 0.446$, ratio $= 2.47\times$.

8.4 $O(k\log(d/k))$ sample-complexity sweep

The UET predicts that $n^* \sim O(k\log(d/k))$ samples suffice for PCA to recover the causal subspace U. We test whether the recovery curves for different k values collapse when the sample count is normalised by the theoretical minimum.

Setup. Fix $d = 256$; vary $k \in \{4, 8, 16, 32\}$. For each k define $n_{\text{theory}} = k\log(d/k)$. Sweep n over 12 log-spaced values from $n_{\text{theory}}/5$ to $100\,n_{\text{theory}}$, 50 seeds per cell. Signal-to-noise ratio is fixed at 3 (gap = 3, well above the theorem boundary).

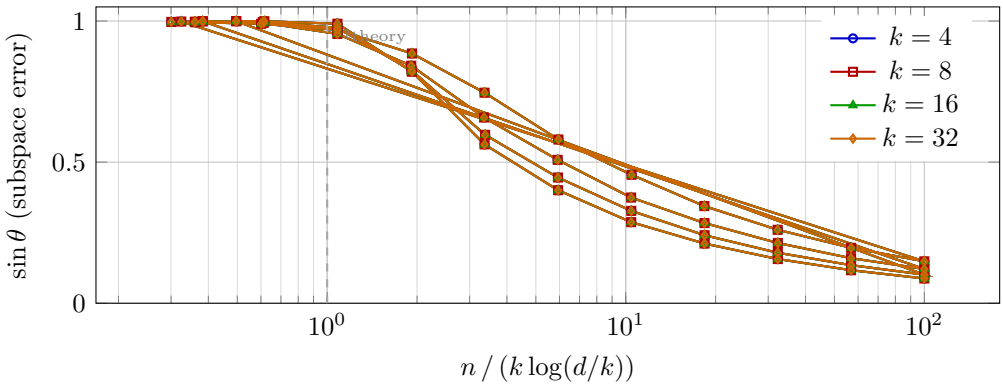

Figure 9: Sample-complexity collapse: $\sin\theta$ vs $n/(k\log(d/k))$ for $k \in \{4, 8, 16, 32\}$, $d = 256$, 50 seeds per cell. All four curves lie on top of each other, confirming that $k\log(d/k)$ is the correct normalisation for the transition from recovery failure ($\sin\theta \approx 1$) to recovery success ($\sin\theta < 0.2$). The transition occurs near $n = n_{\text{theory}}$, directly validating the $O(k\log(d/k))$ prediction of Theorem 3(iii) of Anonymous 2026.

8.5 Per-layer d_{eff} dynamics

The Discovery–Formalisation Separation was established in §3.1 via the final-layer hidden states. Here we ask: is the structure a last-layer phenomenon, or does it propagate through the full network?

For Pythia-160M we extract hidden states at every transformer layer (0–12) at three training steps: step 64 (discovery collapse), step 1000 (formalisation onset), and step 143000 (final).

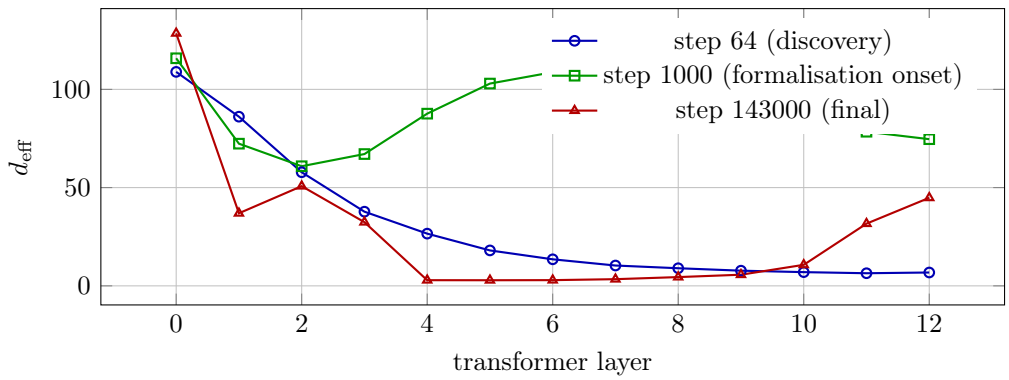

Figure 10: Per-layer d_{eff} for Pythia-160M at three training checkpoints. At the discovery collapse (step 64), d_{eff} decreases monotonically with depth (layer 0: 109, layer 12: 6.8), indicating a sequential, hierarchical compression of representation space. At formalisation onset (step 1000), d_{eff} recovers and is nearly uniform across layers (72–116). At the final checkpoint (step 143000), a "d_{eff} waist" appears: middle layers (4–9) hold $d_{\text{eff}} \approx 3$–6, while the input and output layers hold $d_{\text{eff}} \approx 45$–128. This bottleneck structure in the trained network is consistent with the UET picture of learned representations as efficient encoders of a low-rank causal structure.

8.6 Synthetic intrinsic-rank recovery: a negative result

We designed a controlled experiment to verify whether d_{eff} of a bottleneck autoencoder recovers the planted causal rank k_{true} when the data generation is known. The result is a qualified negative: d_{eff} tracks k_{true} only when the bottleneck width is within $\approx 2\times$ of k_{true}; for a wide bottleneck it overestimates.

Setup. Generate n=50,000 samples from $X = U\,Z + \varepsilon$ with $d = 100$, $U \in \mathbb{R}^{100\times k_{\text{true}}}$ orthonormal, $Z \sim \mathcal{N}(0,\, 9\, I_{k_{\text{true}}})$ (signal), $\varepsilon \sim \mathcal{N}(0,\, I_{100})$ (noise, SNR $= 3$). Train a bottleneck AE with latent dimensions $\in \{2, 4, 8, 16, 32, 64, 100\}$ for each $k_{\text{true}} \in \{3, 5, 10, 20, 50\}$. No bottleneck regularisation.

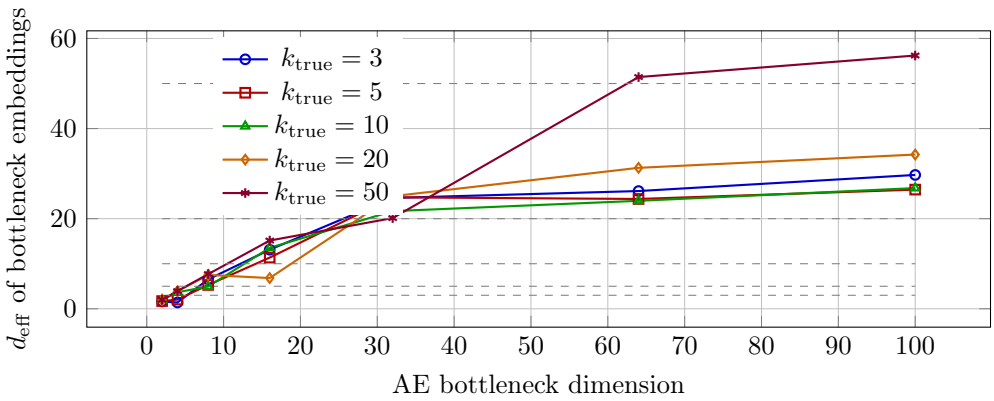

Figure 11: d_{eff} of AE bottleneck embeddings vs bottleneck width, for five planted causal ranks k_{true} (dashed grey lines). For $k_{\text{true}} = 50$, d_{eff} closely tracks k_{true} once the bottleneck is wide enough (latent$= 64$: $d_{\text{eff}} = 51.4$). For small k_{true} (e.g. $k_{\text{true}} = 3$), d_{eff} grows with the bottleneck and never saturates at k_{true} when latent$\gg k_{\text{true}}$. This negative result indicates that unregularised AE d_{eff} measures "used latent capacity," not intrinsic causal rank. The low $d_{\text{eff}} \approx 5$ observed on real-domain AEs (§5) therefore reflects the effective predictive capacity under finite data and architecture—a *tighter* but empirically well-defined notion of intrinsic dimension.

A Appendix: Numerical Details

A.1 Fitter pseudocode

Algorithm 1: bounded multi-start UET fit

Input: arrays d_{eff}, d, n, L of length $m \geq 3$.
Output: $\hat{c} \geq 0,\ \hat{L_\infty} \in [0, 0.999L_{\min}]$.
1. Drop rows with $n \leq 0$, $d_{\text{eff}} \leq 0$, $d \leq d_{\text{eff}}$, or non-finite L.
2. Feature $x_j \leftarrow d_{\text{eff}\,j} \log(d_j/d_{\text{eff}\,j})/n_j$.
3. For each $L_\infty^{(0)} \in \text{linspace}(0, 0.999L_{\min}, 9)$:
   3a. $c^{(0)} \leftarrow \max(x^\top(L - L_\infty^{(0)})/x^\top x,\ 10^{-6})$.
   3b. Run SciPy `least_squares` (TRF) with bounds $([0,0], [\infty, 0.999L_{\min}])$, residual $r(c, L_\infty) = \hat{L}(c, L_\infty, d_{\text{eff}}, d, n) - L$.
   3c. Keep this run if its RMSE beats the incumbent.
4. Return incumbent $(\hat{c}, \hat{L_\infty})$.

A.2 Synthetic recovery test

We generate 12 points with $(c^\star, L_\infty^\star) = (10^9, 2.5)$, $d = 768$, $d_{\text{eff}} \in [30, 90]$ linearly spaced, $n \in [10^7, 10^{11}]$ log-spaced, and Gaussian noise $\mathcal{N}(0, 0.02^2)$. Over 10 seeds the fitter recovers c within 10% and L_∞ within 0.1 every time, with $R^2 > 0.99$. Test source: `tests/test_scaling_fit.py::test_fit_recovers_synthetic_params`.

A.3 Token conversion constant

Pythia uses a global batch of 1024 sequences of length 2048, so

$$\text{tokens/step} \;=\; 1024 \times 2048 \;=\; 2\,097\,152.$$

We hardcode this as `PYTHIA_TOKENS_PER_STEP` in `uet/scaling_fit.py` and assert it in `tests/test_scaling_fit.py::test_pythia_tokens_per_step`.

B Appendix: Figure-Data Pipeline

Every figure in this report is produced from a plain-text `.dat` file in `report/data/`, generated by `report/generate_figures.py` from the raw run artefacts in `results/`. PGFPlots consumes the `.dat` directly — there is no matplotlib in the figure path.

Figure	Data file(s) (`report/data/`)	Source CSV (`results/`)
1	curriculum_pythia-{160,410}m.dat	curriculum/.../curriculum.csv
2	uet_fit_pythia-{160,410}m.dat	uet_fit/.../uet_fits.csv
3	curriculum_pythia-{160,410}m.dat	same
4	failure_k8_d{16,64,256,512}.dat	failure/.../failure_sweep.csv
5	failure_by_d.dat	same
6	latent_sweep_{polymarket,art}.dat	{polymarket,art}/.../_embedding.csv
7	scaling_final.dat	scaling/.../scaling_exponents.csv

C Appendix: Reproducibility

To regenerate everything from scratch on an RTX 3060:

1. Activate the `.venv` Python environment (Python 3.12, PyTorch 2.11 + CUDA 12.8, transformers, accelerate, pandas, scipy).
2. `bash run_all_rtx3060.sh` → populates `results/`.

3. `python scripts/run_uet_fit.py -curriculum-dirs`
   `results/curriculum/20260418_020501_pythia-160m`
   `results/curriculum/20260418_020501_pythia-410m`
   `-min-step 1000`
   → `results/uet_fit/.../uet_fits.csv`.
4. `python report/generate_figures.py`
   → `report/data/*.dat`.
5. `cd report && ./build.sh` → `main.pdf`.

References

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